09

Future of robotics

The future of robotics covers the advancement of robotics capable of performing tasks autonomously or semi-autonomously, adapting to new, real-life inputs with increasing degrees of autonomy and dexterity, including autonomous mobile robots and humanoid robots.

The trend—and why it matters

Over the past six decades, robots have become a familiar presence in advanced manufacturing. Today, more than four million industrial robots work in settings such as automobile plants. Interest in physical robotics, enabled by advances in AI, surged beyond industrial settings in 2024. A growing number of companies are developing robots with different form factors, from robotic arms and quadrupeds to humanoid machines that navigate and operate in physical spaces designed for humans. The same kinds of AI foundation models that enable the creation of chatbots are being trained to control robots to respond flexibly in novel situations. The combination of new form factors and more flexible control systems could put more multipurpose or even general-purpose robots to work.

Robots are increasingly being designed and deployed in environments beyond manufacturing, including airports, large-format stores, and restaurants. While annual industrial-robot installations are forecast to grow in the mid-single-digit percentages annually for the next few years, with half of the new installations in China, the market for service robots in areas such as logistics, hospitality, and agriculture has been growing many times faster—20 to 35 percent annually. Although interest in "humanoid" robots is growing, many robots do not resemble humans at all. For example, autonomous mobile robots (AMRs) that transport materials within a warehouse resemble a robotic household vacuum more than a person. An increasing number of robots are being designed to operate safely side by side with human workers. Known as cobots, short for collaborative robots, they expand the number of tasks and situations in which robots can be deployed beyond industrial settings, where safety walls often separate them from human workers.

Despite significant advancements, challenges persist. Untethered humanoid robots are constrained by power: A state-of-the-art humanoid robot can operate for about four hours before needing to be recharged, which takes roughly two hours. Humanoid robots also fall down—the balance needed to stay atop two feet is harder to achieve than one might expect. Handlike devices are often much slower than human hands when it comes to grasping objects, and other devices, such as suction cups, can be more effective for certain tasks. Other critical priorities to address are workforce reskilling, physical safety of human coworkers, and robust cybersecurity frameworks.[1]

[1] Future of robotics insights provided by the Stanford Robotics Center.

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Scoring the trend

Compared with other trends, robotics has a smaller overall footprint. Nonetheless, interest and innovation metrics for robotics increased by double digits from 2023 to 2024. Media attention to humanoid robots partly explains the increased interest, but real technological innovations are being incorporated into robots, including equipping them with AI-driven capabilities like natural-language programming and advanced sensing. The market for robots is also expanding, with adoption in emerging sectors like agriculture and healthcare, the development of cobots, and broader deployment in manufacturing and logistics.

Equity investment, 2024

$7 billion

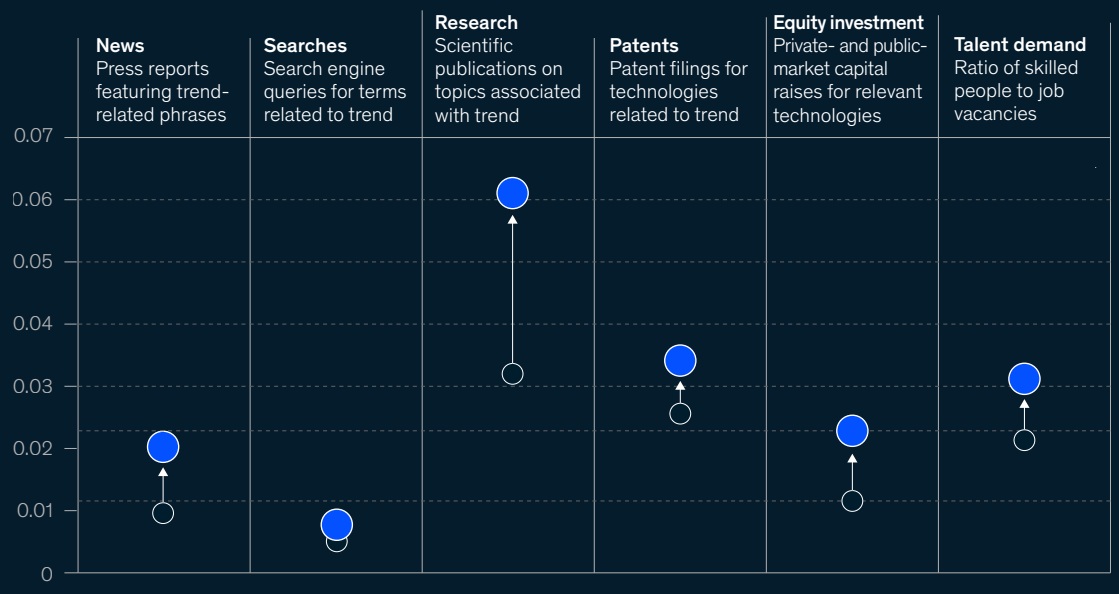
Job postings, 2023–24, % difference

–2%

Score, by vector (0 = lower; 1 = higher)

○ 2020 ● 2024

1 — 0 — *Range shown*



Note: For each vector, we used a defined set of data sources to find occurrences of keywords associated with each of the 13 trends, screened those occurrences for valid mentions of activity, and indexed the resulting numbers of mentions on a 0–1 scoring scale that is relative to the trends studied.

Latest developments

Recent developments involving robotics include foundation models for robotics and humanoid robots:

— *An increasing number of robot developers are building "foundation models" to control robots in the physical world—the same AI solution that underpins progress in the virtual world.* These models are giant, simulated neural networks trained on immense quantities of varied unstructured data to generate unstructured output data, for example, large language models (LLMs). In the context of robotics, the goal is for foundation models to generate robot behaviors rather than language. Additionally, the industry hopes that, like LLMs, foundation models for robots will be able to generalize from their training data and generate appropriate responses to a wide range of inputs, even those they were not explicitly trained on.

— *The volume of training data is a limiting factor for using foundation models to control robots.* LLMs are trained on all the language data on the internet, but finding enough data to train robot foundation models on how to move is not easy. Researchers are getting creative by collaborating to collect bigger real-world

training data sets, creating simulated worlds, teaching robots to watch videos, and paying workers to perform tasks repeatedly while being recorded, among other efforts.

— *Multiple companies are developing humanoid robots, which have vaguely human shapes—two arms, two legs, a body, and sometimes a head.* The practical reason for developing robots with human forms is to deploy them in physical spaces designed for human beings; after all, legs are more suitable than wheels for climbing stairs. These robots are designed for flexible applications, as they can be deployed for multipurpose or even general-purpose use.

— *Robotics has expanded into more sectors, particularly into services, including cobots working safely alongside human workers.* For example, robots are now working in the food service sector, preparing food. The healthcare sector has also expanded its use of robots, using them to deliver items in hospitals.

'The moment for AI-driven robotics has arrived. In the face of labor shortages and escalating production costs, the technology will soon become a dimension of competitive differentiation and a CEO-level agenda topic. The market opportunity is massive—up to approximately $900 billion by 2040—but adoption still requires addressing challenges around capability development, operating model, IT/OT [information technology/operational technology] infrastructure, and a playbook for automation at scale.'

— Ani Kelkar, partner, Boston

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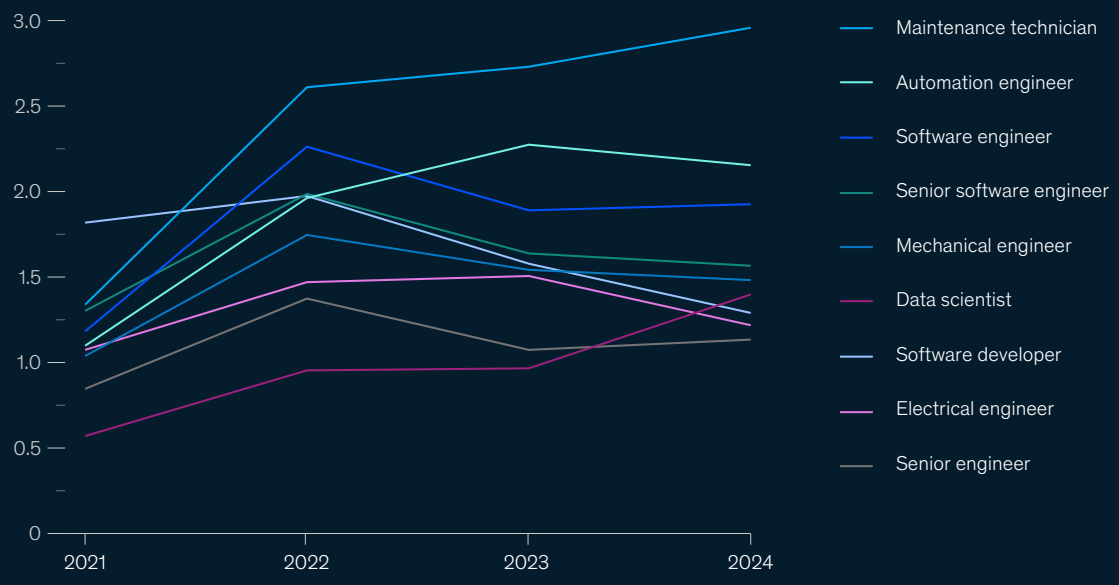
Demand

Although the number of jobs in robotics remains small compared with other sectors, demand increased across almost all job titles between 2021 and 2024. Positions such as maintenance technician, data scientist, and automation engineer had especially strong growth, reflecting expanded automation needs in manufacturing, logistics, and healthcare that require skilled technicians; the integration of AI systems demanding more data scientists to develop machine learning (ML) models for robotic decision-making; and the adoption of technologies necessitating automation engineers to implement cobots and Internet of Things–connected systems.

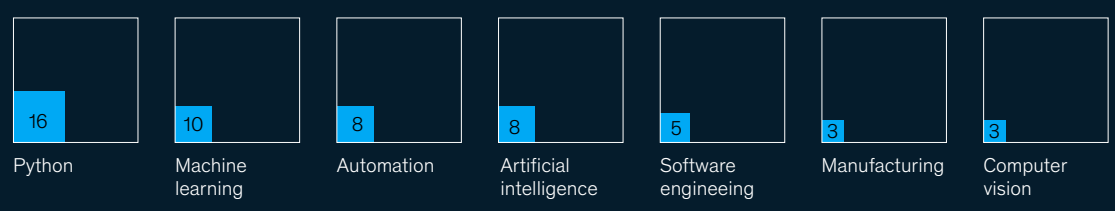
Job postings, by title, 2021–24, thousands



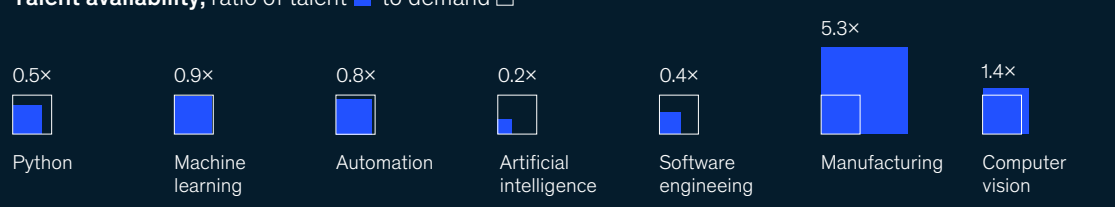
Skills availability

As robotics increasingly incorporates AI-powered systems, the demand for expertise in ML, AI, automation, and computer vision is growing, yet the talent pool remains constrained. Companies will need to bridge these skill gaps to fully unlock robotics technologies' transformative potential.

Talent required, % share of postings requiring skill



Talent availability, ratio of talent to demand



Adoption developments across the globe

Adoption score: 2—Experimentation. Organizations are testing the functionality and viability of the technology using small-scale prototypes, typically without a focus on a near-term ROI. Few companies outside of manufacturing and e-commerce are scaling or have fully scaled the technology.

Although adoption varies across robotic technologies, several piloting initiatives are underway in warehouses, manufacturing, and general-purpose use cases.

In real life

Real-world examples involving the use of foundation models in robotics include the following:

— *Covariant introduced RFM-1, a robotics foundation model that gives robots human-like reasoning.* This model helps robots understand how objects move and interact in the real world, follow language-based instructions, and reflect on their own actions.[2]

— *Figure AI introduced Helix, a vision-language-action (VLA) model that enables humanoid robots to perform complex tasks, such as putting away groceries.* According to Figure AI, Helix is a step toward adaptable, real-world robotic applications, enabling dynamic object recognition and collaboration without prior training.[3]

Real-world examples involving the development of humanoid robots include the following:

— *Boston Dynamics introduced Electric Atlas, a fully electric humanoid robot with enhanced strength and a range of motion that can handle heavy objects in industrial settings.* Hyundai plans to deploy Electric Atlas in automotive manufacturing as a cobot.[4]

— *Tesla's Optimus is a humanoid robot designed for general-purpose tasks in industrial and domestic environments.* Standing five feet, eight inches tall and weighing 125 pounds, Optimus is equipped with advanced AI and 22 degrees of freedom in its hands. The company envisions deploying Optimus in various sectors, including manufacturing, healthcare, and home services.[5]

— *GXO Logistics has a multiyear agreement with Agility Robotics to deploy the Digit humanoid robot in its facilities, the first industry agreement of its kind.* Digit will automate tasks like moving totes and managing palletization, which can enhance operational efficiency and safety. Digit has performed successfully in a GXO pilot program, maximizing capacity and keeping up with demand in sectors like manufacturing, third-party logistics, retail, and e-commerce.[6]

Real-world examples involving expanded industry applications of robotics include the following:

— *Amazon has deployed AI-driven cobots across its warehouses to automate both pick-and-place and palletization tasks, significantly boosting throughput and accuracy.* These robots can handle a wide range of items, leveraging force sensors and AI-powered vision systems. This combination enables them to pick up and place objects with dexterity and tactile sensitivity.[7]

— *Launched in October 2023, BotBob is KFoodtech's advanced-robotic-kitchen solution that addresses labor shortages and operational costs in restaurants.* BotBob can cook up to six servings of traditional Korean stew in just 3.5 minutes, improving efficiency and reducing wait times.[8]

— *Tortuga AgTech's robots can identify and pick ripe fruit with 98 percent accuracy, requiring only one human supervisor.* These robots address labor shortages and reduce crop damage during harvesting.[9]

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2 Andrew Sohn et al., "Introducing RFM-1: Giving robots human-like reasoning capabilities," Covariant, March 11, 2024.
3 "Helix: A vision-language-action model for generalist humanoid control," Figure, February 20, 2025.
4 "An electric new era for Atlas," Boston Dynamics, April 13, 2024.
5 Niamh Ancell, "Optimus' new hand upgrade includes 22 degrees of freedom: What does it mean?," Cybernews, November 29, 2024; Jijo Malayil, "Watch: Tesla Optimus robot catches high-speed tennis balls with new hand upgrade," Interesting Engineering, November 29, 2024.
6 "GXO signs industry-first multi-year agreement," GXO Logistics press release, June 27, 2024.
7 "How AI-powered cobots transform the logistics industry," Techman Robot, accessed March 26, 2025; Dominic Preston, "Amazon develops a robot that 'feels' touch, just like its human workers," *Verge*, May 7, 2025.
8 Daehyun Song, "KFoodtech Inc. introduces BotBob – a next generation robotic kitchen solution for restaurant industry," KoreaTechDesk, September 13, 2023.
9 Jennifer Strailey, "Oishii acquires robotics company Tortuga AgTech, extends harvesting capabilities," *Packer*, March 24, 2025.

— *Sanctuary AI's advanced haptics tactile sensors*[10] *and Meta AI's*[11] *Digit 360 improved robotic dexterity to enable tasks like blind picking and slippage detection.* Digit 360 can detect spatial details as fine as seven microns and forces as small as one millinewton, which provides a foundation for more versatile and sensitive robotic applications across industries.

Underlying technologies

The technologies that power robotics include the following:

— *Advanced AI and ML.* These sophisticated algorithms and models enable robots to learn, adapt, and make complex decisions based on data and experience.

— *Sensor and vision systems.* Robotic sensing and vision technologies are hardware and software combinations that allow robots to perceive and interpret their environment, enabling tasks like object recognition and navigation.

— *Advanced actuators and motion control.* These are components and systems that precisely control robot movements, enabling complex and accurate physical interactions.

— *Human–robot collaboration.* Collaborative technologies allow robots to work safely and effectively alongside humans, enhancing productivity in various industries.

— *Robotics as a service (RaaS).* RaaS is a business model that provides robotic solutions on a subscription or pay-per-use basis, increasing access to advanced robotics for more businesses.

— *Autonomous navigation and decision-making.* These capabilities allow robots to move and operate independently in various environments and make real-time decisions based on their surroundings.

— *Tactile sensing.* Sensing technologies enable robots to detect and respond to physical contact, improving their ability to handle objects and interact with their environment.

— *Purpose-built batteries.* These batteries are specialized power sources designed to meet the specific energy requirements and operational demands of robotic systems.

Key uncertainties

The major uncertainties affecting the future of robotics include the following:

— *Uncertainties about the risk and trust framework for robotics emerge as they mix into our labor force beyond the traditional manufacturing use cases.* For example, the liability and safety protocols for autonomous robots may need to be redefined, as existing frameworks have difficulty addressing AI-driven decision-making. Establishing international standards for human–robot collaboration safety remains an important challenge.

— *The fascination, fear, and uncertainty surrounding humanoid robots have given rise to concerns that they may develop to meet and exceed the capabilities of certain employees.* As robots take on more advanced tasks, the nature of human work is evolving. For example, autonomous robots are increasingly being used to handle tasks like room service and cleaning in hospitality. While these systems improve efficiency, they also spur public fears about job displacement.

Big questions about the future

Businesses and leaders may want to consider a few questions when moving forward with robotics:

— When does it make strategic and financial sense to retrofit robotics into existing brownfield environments versus starting from a clean slate?

— Which untapped industries or service sectors are poised for a breakthrough in robotics implementation?

— Where do cobots create real value—and where do they get in the way?

— What new ethical and regulatory challenges will emerge with the rapid advancement of AI in robotics?

— What will it take to build employee and customer trust in robotics?

[10] "Sanctuary AI new tactile sensors enable richer sense of touch," Sanctuary AI press release, December 19, 2024.
[11] Chris McKay, "Meta is developing a robot hand that can touch and feel," Maginative, October 31, 2024.